

A Two-Dimensional Super-Resolution Method for Reconfigurable Intelligent Surface-Assisted Near-Field Localization

Feng Xi

*Department of Electronic Engineering
Nanjing University of Science and Technology
Nanjing 210094, China*

Dehui Yang

*School of Mathematics and Physics
Xi'an Jiaotong-Liverpool University
Suzhou 215123, China*

Abstract—Reconfigurable intelligent surface (RIS)-aided localization in the radiating near-field requires range-aware spherical-wave models, which inherently couple angles and ranges and thus complicate accurate 3D positioning. Using the Fresnel approximation, we show that the RIS response can be expressed as the element-wise product of a 2D far-field steering vector and a range-dependent quadratic-phase chirp. By modeling these chirp components within a low-dimensional subspace, we reformulate the joint recovery of azimuth, elevation, and range under a 2D super-resolution framework, resulting in a 2D atomic norm minimization (2D-ANM) problem. Solving this via semi-definite programming (SDP) yields gridless azimuth-elevation estimation and high-accuracy range recovery. Simulations demonstrate accurate 3D localization and enhanced robustness of the proposed scheme compared with subspace and compressive sensing methods.

Index Terms—Reconfigurable intelligent surface, near-field localization, super-resolution, atomic norm minimization, joint azimuth-elevation-range estimation

I. INTRODUCTION

Reconfigurable intelligent surfaces (RISs) are low-power, nearly passive metasurfaces that program element-wise phases to reshape radio propagation, enabling robust connectivity and high-accuracy localization even under non-line-of-sight conditions [1], [2]. As apertures scale and carriers move to mmWave/THz bands, users increasingly operate within the radiating near-field (NF) region of large RISs, where wavefronts are spherical and range-dependent. In this regime, conventional far-field (FF) plane-wave models become inaccurate. While the spherical-wave model supports 3D positioning with a uniform planar array (UPA) RIS [3]–[5], joint estimation of azimuth, elevation, and range is challenging due to the high dimensionality of the parameter space and strong nonlinear coupling among these parameters.

Radio localization is central to array processing and a key enabling technology for integrated sensing and communication (ISAC) in 6G [6]. Two representative approaches are subspace [7] and compressive sensing (CS) methods [8]. Subspace approaches rely on accurate sample-covariance estimation and

typically require many snapshots alongside multi-dimensional search. These requirements are difficult to satisfy in dynamic scenarios with short channel coherence time, where accurate localization must be accomplished using only a few snapshots to reduce latency and pilot overhead. CS approaches discretize continuous parameters to obtain a sparse linear model, but suffer from grid mismatch and resolution loss. Furthermore, multi-dimensional parameter estimation often entails constructing high-dimensional dictionaries, which substantially increases storage and computational burdens. In RIS-assisted localization, integrating a very large number of low-cost elements on a single panel places user equipment (UE) frequently in the RIS's NF region, necessitating a shift from the FF planar-wave to the NF spherical-wave model. Meanwhile, RISs provide only reflective control with limited on-board processing, further complicating accurate NF localization.

To address these issues, both subspace and CS methods have been tailored to the RIS-aided scenarios. For subspace methods, [4] constructs a down-sampled Toeplitz covariance to partially decouple parameters, and [9] exploits array symmetry with spatial smoothing to obtain a well-conditioned covariance matrix. For CS methods, [10]–[12] design NF codebooks to facilitate orthogonal matching pursuit (OMP)-based sparse recovery, while [5] combines OMP with maximum likelihood (ML) refinement to improve accuracy. Despite these advances, achieving high-resolution multi-dimensional estimation from few snapshots remains difficult due to parameter coupling.

This paper develops a super-resolution framework for RIS-assisted NF localization by exploiting the Fresnel approximation and the inherent structure of the UE-RIS channel. Under the Fresnel model, the RIS response can be expressed as a 2D FF steering vector modulated by a range-dependent quadratic-phase chirp. For a UPA RIS, the response is further separable across the horizontal and vertical axes via a Kronecker structure. Leveraging these properties and modeling the chirps with low-dimensional subspaces, we reformulate the RIS-assisted NF localization as a 2D super-resolution problem [13] that simultaneously recovers the 2D FF steering vectors and the unknown chirp components. As a result, this problem can be solved via 2D atomic norm minimization

The work of F. Xi was supported in part by the National Natural Science Foundation of China under grant No. 62471230. The work of D. Yang was supported in part by the Research Development Fund (RDF-23-02-080) from Xi'an Jiaotong-Liverpool University.

(2D-ANM), yielding gridless azimuth-elevation estimation and high-accuracy range recovery. Notably, although the Fresnel approximation has been adopted in prior studies [4], [10], [11], our approach treats the range-dependent quadratic-phase terms as unknown waveforms that modulate FF steering vectors. This viewpoint enables high-precision 3D NF localization via a gridless 2D super-resolution formulation and is fundamentally distinct from conventional dictionary-based or search-based treatments that parameterize range explicitly on a grid or via multi-dimensional refinement. In this sense, by establishing a Kronecker-structured Fresnel factorization, the proposed method extends classical 2D super-resolution frameworks [14], [15] from the FF plane-wave setting to the NF spherical-wave regime, in which the spherical-wave manifold can be cast as a multiple measurement vector (MMV) 2D line-spectral model. Simulations show that the proposed method achieves superior performance over subspace and CS methods, particularly in regimes with limited snapshots.

II. SIGNAL MODEL

We consider a RIS-aided localization system where a base station (BS) equipped with M antennas communicates with K single-antenna UEs through an N -element RIS. The RIS forms a UPA with N_H elements per row and N_V elements per column ($N = N_H N_V$). Both N_H and N_V are assumed to be odd integers, and the element at the geometric center serves as the reference. The RIS lies on the y - z plane with its array axes aligned with the y - and z -directions. Let d_H and d_V be the inter-element spacings along the y - and z -axes, respectively. We index the RIS elements by (n_H, n_V) with $n_H \in \mathcal{N}_H = \{-\frac{N_H-1}{2}, \dots, \frac{N_H-1}{2}\}$, $n_V \in \mathcal{N}_V = \{-\frac{N_V-1}{2}, \dots, \frac{N_V-1}{2}\}$, and map (n_H, n_V) to the scalar index $n = (n_H + \frac{N_H-1}{2})N_V + (n_V + \frac{N_V-1}{2}) + 1$. For notational simplicity, we use n and (n_H, n_V) interchangeably hereafter.

The system aims to localize the K UEs, which operate in the radiating NF of the RIS. We consider the scenario where direct UE-BS links are blocked, so both uplink and downlink transmissions occur via the RIS. The spherical coordinate of the k th UE is (r_k, ψ_k, θ_k) , i.e., its Cartesian coordinate $\mathbf{p}_k = r_k(\cos(\psi_k)\cos(\theta_k), \sin(\psi_k)\cos(\theta_k), \sin(\theta_k))$, where r_k is the distance from the k th UE to the reference RIS element, and (ψ_k, θ_k) are the azimuth and elevation angles of arrival (AOAs) at the RIS.

Denote the RIS-BS channel as $\mathbf{H} \in \mathbb{C}^{M \times N}$, which is assumed to be static and known since the BS and the RIS are stationary. Then the received signal at the BS during the t th time-slot is

$$\mathbf{y}(t) = \sum_{k=1}^K \sqrt{p_k} \mathbf{H} \Phi(t) \mathbf{g}_k s_k(t) + \mathbf{w}(t), \quad (1)$$

where p_k is the transmit power of the k th UE, $s_k(t)$ is the pilot symbol, $\mathbf{w}(t)$ is the additive random noise, and $\Phi(t) = \text{diag}(e^{j\phi_1(t)}, \dots, e^{j\phi_N(t)})$ is the RIS reflection matrix with $\phi_n(t)$ denoting the phase shift applied by the n th RIS element in the time-slot t . The UE-RIS channel is $\mathbf{g}_k = \eta_k \mathbf{b}(\psi_k, \theta_k, r_k) \in \mathbb{C}^N$, with path loss η_k and RIS response

$\mathbf{b}(\psi_k, \theta_k, r_k)$. Let $r_k^{(n)}$ denote the distance from the k th UE to the n th RIS element. The n th element of $\mathbf{b}(\psi_k, \theta_k, r_k)$ is given by

$$[\mathbf{b}(\psi_k, \theta_k, r_k)]_n = \exp \left\{ j \frac{2\pi}{\lambda} (r_k - r_k^{(n)}) \right\}, \quad (2)$$

where λ is the carrier wavelength.

In the radiating NF region, $r_k^{(n)}$ admits the Fresnel approximation:

$$r_k^{(n)} \approx r_k - n_H d_H \sin(\psi_k) \cos(\theta_k) - n_V d_V \sin(\theta_k) + \frac{n_H^2 d_H^2 + n_V^2 d_V^2}{2r_k}. \quad (3)$$

Consequently, (2) can be approximated as

$$[\mathbf{b}(\psi_k, \theta_k, r_k)]_n \approx \exp \left\{ j \left((n_H \alpha_k + n_V \beta_k) - (n_H^2 d_H^2 + n_V^2 d_V^2) \gamma_k \right) \right\}, \quad (4)$$

with spatial frequencies $\alpha_k = \frac{2\pi}{\lambda} d_H \sin(\psi_k) \cos(\theta_k)$, $\beta_k = \frac{2\pi}{\lambda} d_V \sin(\theta_k)$, and $\gamma_k = \frac{\pi}{\lambda r_k}$.

Define the FF steering vectors $\mathbf{a}_H(\alpha)$, $\mathbf{a}_V(\beta)$ and the quadratic-phase chirp vectors $\mathbf{q}_H(\delta)$, $\mathbf{q}_V(\delta)$ as

$$\begin{aligned} \mathbf{a}_H(\alpha) &= [e^{jn_H \alpha}]_{n_H \in \mathcal{N}_H}, & \mathbf{a}_V(\beta) &= [e^{jn_V \beta}]_{n_V \in \mathcal{N}_V}, \\ \mathbf{q}_H(\delta) &= [e^{-jn_H^2 \delta}]_{n_H \in \mathcal{N}_H}, & \mathbf{q}_V(\delta) &= [e^{-jn_V^2 \delta}]_{n_V \in \mathcal{N}_V}. \end{aligned}$$

The RIS response can be compactly rewritten as

$$\begin{aligned} \mathbf{b}(\psi_k, \theta_k, r_k) &\approx (\mathbf{a}_H(\alpha_k) \otimes \mathbf{a}_V(\beta_k)) \odot (\mathbf{q}_H(d_H^2 \gamma_k) \otimes \mathbf{q}_V(d_V^2 \gamma_k)) \\ &= (\mathbf{a}_H(\alpha_k) \odot \mathbf{q}_H(d_H^2 \gamma_k)) \otimes (\mathbf{a}_V(\beta_k) \odot \mathbf{q}_V(d_V^2 \gamma_k)), \end{aligned} \quad (5)$$

where $\otimes$ and $\odot$ denote the Kronecker and Hadamard products, respectively. Thus, estimating the original parameters $\{\psi_k, \theta_k, r_k\}_{k=1}^K$ is equivalent to recovering the parameters $\{\alpha_k, \beta_k, \gamma_k\}_{k=1}^K$. Expression (5) reveals a Kronecker-structured Fresnel approximation that decouples the horizontal and vertical array responses. This separability is essential for formulating the 2D atomic norm and deriving the joint azimuth-elevation-range estimation method.

III. TWO-DIMENSIONAL SUPER-RESOLUTION FRAMEWORK

In this section, we exploit the Kronecker-structured Fresnel approximation in (5) to reformulate the RIS-assisted NF localization problem under a 2D super-resolution framework and solve it via 2D-ANM.

Following [16], we approximate the quadratic-phase terms by low-dimensional subspaces, i.e.,

$$\mathbf{q}_H(d_H^2 \gamma_k) \approx \mathbf{B}_H \mathbf{u}_H^k, \quad \mathbf{q}_V(d_V^2 \gamma_k) \approx \mathbf{B}_V \mathbf{u}_V^k, \quad (6)$$

where $\mathbf{B}_H \in \mathbb{C}^{N_H \times J_H}$, $\mathbf{B}_V \in \mathbb{C}^{N_V \times J_V}$ are subspace matrices with $J_H \ll N_H$ and $J_V \ll N_V$, and $\mathbf{u}_H^k$ and $\mathbf{u}_V^k$ are the corresponding coefficients. The subspaces can be constructed, for example, by discretizing the quadratic phase [16], [17] or using discrete prolate spheroidal sequences (DPSSs) [18], [19].

Let $(\mathbf{b}_H^{(n_H)})^H$, $(\mathbf{b}_V^{(n_V)})^H$ be the n_H th, n_V th rows of $\mathbf{B}_H$ and $\mathbf{B}_V$, respectively, and $\mathbf{e}_H^{(n_H)}$, $\mathbf{e}_V^{(n_V)}$ be the corresponding

canonical basis vectors. Define the linear operator $\mathcal{P}(\cdot) : \mathbb{C}^{J_H J_V \times N} \mapsto \mathbb{C}^N$ with its n th element given by

$$[\mathcal{P}(\mathbf{Z})]_n = \langle \mathbf{Z}, \mathbf{b}_H^{(n_H)} (\mathbf{e}_H^{(n_H)})^H \otimes \mathbf{b}_V^{(n_V)} (\mathbf{e}_V^{(n_V)})^H \rangle, \quad (7)$$

where $\langle \mathbf{A}, \mathbf{B} \rangle = \text{Tr}(\mathbf{B}^H \mathbf{A})$. Using $\langle \mathbf{A} \otimes \mathbf{B}, \mathbf{C} \otimes \mathbf{D} \rangle = \langle \mathbf{A}, \mathbf{C} \rangle \langle \mathbf{B}, \mathbf{D} \rangle$, the RIS response (5) becomes

$$\mathbf{b}(\psi_k, \theta_k, r_k) = \mathcal{P}(\mathbf{u}_H^k \mathbf{a}_H^T(\alpha_k) \otimes \mathbf{u}_V^k \mathbf{a}_V^T(\beta_k)). \quad (8)$$

Define the rank- K matrix

$$\begin{aligned} \mathbf{X}_o &= \sum_{k=1}^K \tilde{\eta}_k \mathbf{u}_H^k \mathbf{a}_H^T(\alpha_k) \otimes \mathbf{u}_V^k \mathbf{a}_V^T(\beta_k) \\ &= \sum_{k=1}^K \tilde{\eta}_k (\mathbf{u}_H^k \otimes \mathbf{u}_V^k) (\mathbf{a}_H(\alpha_k) \otimes \mathbf{a}_V(\beta_k))^T, \end{aligned} \quad (9)$$

with $\tilde{\eta}_k = \sqrt{p_k} \eta_k$. Without loss of generality, set the pilot symbols $s_k(t) = 1$ for $t = 1, \dots, L$. Stacking $\mathbf{y} = [\mathbf{y}^T(1), \dots, \mathbf{y}^T(L)]^T$ yields

$$\mathbf{y} = \bar{\mathbf{H}} \mathcal{P}(\mathbf{X}_o) + \mathbf{w}, \quad (10)$$

where $\bar{\mathbf{H}} = [(\mathbf{H}\Phi(1))^T, \dots, (\mathbf{H}\Phi(L))^T]^T$ and $\mathbf{w} = [\mathbf{w}^T(1), \dots, \mathbf{w}^T(L)]^T$. Therefore, our problem reduces to recovering the low-rank matrix $\mathbf{X}_o$ from $\mathbf{y}$. Moreover, due to the Kronecker-structured steering vector in (9), $\mathbf{X}_o$ follows an MMV 2D line-spectral model, which motivates a gridless 2D-ANM formulation.

Let $\mathbf{d}(\alpha, \beta) = \mathbf{a}_H(\alpha) \otimes \mathbf{a}_V(\beta)$ and define the atomic set $\mathcal{A} = \{\mathbf{u} \mathbf{d}^T(\alpha, \beta) \mid \|\mathbf{u}\|_2 = 1, \mathbf{u} \in \mathbb{C}^{J_H J_V}, \alpha, \beta \in [-\pi, \pi]\}$. The associated atomic norm is

$$\|\mathbf{X}\|_{\mathcal{A}} = \inf_{\eta_l, \alpha_l, \beta_l, \mathbf{u}_l} \left\{ \sum_l |\eta_l| : \mathbf{X} = \sum_l \eta_l \mathbf{u}_l \mathbf{d}^T(\alpha_l, \beta_l) \right\}.$$

Therefore, $\mathbf{X}_o$ is recovered via solving the following convex program:

$$\min_{\mathbf{X}} \|\mathbf{X}\|_{\mathcal{A}} \quad \text{s.t.} \quad \|\mathbf{y} - \bar{\mathbf{H}} \mathcal{P}(\mathbf{X})\|_2 \leq \epsilon, \quad (11)$$

where $\epsilon > 0$ is a predefined parameter controlling the noise level. Problem (11) can be cast as semidefinite programming (SDP) by exploiting the 2-D structure induced by $\mathbf{d}(\alpha_k, \beta_k)$.

Define the two-fold Toeplitz matrix $\text{Toep}(\mathbf{T}) \in \mathbb{C}^{N \times N}$ with a given $\mathbf{T} \in \mathbb{C}^{(2N_H-1) \times (2N_V-1)}$:

$$\text{Toep}(\mathbf{T}) = \begin{bmatrix} \mathbf{T}_0 & \mathbf{T}_{-1} & \cdots & \mathbf{T}_{-(N_H-1)} \\ \mathbf{T}_1 & \mathbf{T}_0 & \cdots & \mathbf{T}_{-(N_H-2)} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{T}_{N_H-1} & \mathbf{T}_{N_H-2} & \cdots & \mathbf{T}_0 \end{bmatrix}, \quad (12)$$

where each block $\mathbf{T}_\ell$ in $\text{Toep}(\mathbf{T})$ is an $N_V \times N_V$ Toeplitz matrix formed by the ℓ th row of $\mathbf{T}$ with $\mathbf{T}_\ell[i, j] = \mathbf{T}[\ell, i - j]$, $0 \leq i, j \leq N_V - 1$.

According to [14], (11) admits an approximate SDP characterization:

$$\begin{aligned} \min_{\mathbf{X}, \mathbf{T}, \mathbf{Q}} \quad & \frac{1}{2} \text{Tr}(\mathbf{Q}) + \frac{1}{2N} \text{Tr}(\text{Toep}(\mathbf{T})) \\ \text{s.t.} \quad & \|\mathbf{y} - \bar{\mathbf{H}} \mathcal{P}(\mathbf{X})\|_2 \leq \epsilon, \\ & \begin{bmatrix} \mathbf{Q} & \mathbf{X} \\ \mathbf{X}^H & \text{Toep}(\mathbf{T}) \end{bmatrix} \succeq 0, \end{aligned} \quad (13)$$

where $\mathbf{Q} \in \mathbb{C}^{J_H J_V \times J_H J_V}$. The SDP can be solved using the off-the-shelf CVX toolbox [20]. Following [21], the computational complexity of solving (13) via interior-point methods (e.g., CVX/SDPT3) scales as $\mathcal{O}((N J_H J_V)^2 (N + J_H J_V)^{2.5})$. For large-scale instances, the alternating direction method of multipliers (ADMM) [22] can be employed to reduce the complexity to $\mathcal{O}(K_{\text{1st}}(N + J_H J_V)^3)$, where K_{1st} denotes the number of ADMM iterations. Moreover, as reported in [23], the complexity of solving (13) can be further reduced by adopting a coordinate-descent strategy.

IV. JOINT AZIMUTH-ELEVATION-RANGE ESTIMATION

Let $(\hat{\mathbf{X}}, \hat{\mathbf{T}}, \hat{\mathbf{Q}})$ denote the optimal solution to the SDP in (13). We now introduce the detailed steps to recover the set of parameters $\{\eta_k, \psi_k, \theta_k, r_k\}_{k=1}^K$ from $(\hat{\mathbf{X}}, \hat{\mathbf{T}}, \hat{\mathbf{Q}})$.

1) Azimuth-elevation estimation: With $\hat{\mathbf{T}}$, we form the two-fold Toeplitz matrix $\hat{\mathbf{R}} = \text{Toep}(\hat{\mathbf{T}}) \in \mathbb{C}^{N \times N}$. By the multilevel Vandermonde decomposition [24], $\hat{\mathbf{R}}$ admits the following Vandermonde decomposition:

$$\hat{\mathbf{R}} = \mathbf{D} \text{diag}([c_1, \dots, c_K]) \mathbf{D}^H, \quad c_k > 0, \quad (14)$$

where $\mathbf{D} = [\mathbf{d}(\hat{\alpha}_1, \hat{\beta}_1), \dots, \mathbf{d}(\hat{\alpha}_K, \hat{\beta}_K)] \in \mathbb{C}^{N \times K}$ is a Vandermonde matrix. The spatial frequencies $\{\hat{\alpha}_k, \hat{\beta}_k\}$ are then extracted from $\hat{\mathbf{R}}$ via the matrix pencil and pairing (MaPP) procedure [24], which are then mapped to angles via $\hat{\theta}_k = \arcsin\left(\frac{\hat{\beta}_k \lambda}{2\pi d_V}\right)$ and $\hat{\psi}_k = \arcsin\left(\frac{\hat{\alpha}_k \lambda}{2\pi d_H \cos \hat{\theta}_k}\right)$.

2) Range estimation: From (9), $\mathbf{X}_o$ can be rewritten as $\mathbf{X}_o = \mathbf{U} \mathbf{D}^T$, where $\mathbf{U} = [\tilde{\eta}_1 (\mathbf{u}_H^1 \otimes \mathbf{u}_V^1), \dots, \tilde{\eta}_K (\mathbf{u}_H^K \otimes \mathbf{u}_V^K)] \in \mathbb{C}^{J_H J_V \times K}$. Projecting $\hat{\mathbf{X}}$ onto $\text{span}(\mathbf{D})$ gives the estimate

$$\hat{\mathbf{U}} = \hat{\mathbf{X}} (\mathbf{D}^T)^\dagger \in \mathbb{C}^{J_H J_V \times K}, \quad (15)$$

where $(\cdot)^\dagger$ denotes the Moore-Penrose inverse. Each column of $\hat{\mathbf{U}}$, denoted as $\hat{\mathbf{u}}_k$, approximates the separable Kronecker form $\mathbf{u}_H^k \otimes \mathbf{u}_V^k$ up to a scaling factor due to the unknown $\tilde{\eta}_k$. Reshaping $\hat{\mathbf{u}}_k$ into a $J_H \times J_V$ matrix and taking its best rank-one approximation $\hat{\mathbf{u}}_k \approx \sigma_k \hat{\mathbf{u}}_H^k (\hat{\mathbf{u}}_V^k)^H$ yields the estimates of $\mathbf{u}_H^k$ and $\mathbf{u}_V^k$ in (6). Then we can reconstruct the quadratic-phase chirp vectors:

$$\hat{\mathbf{q}}_H^k = \mathbf{B}_H \hat{\mathbf{u}}_H^k, \quad \hat{\mathbf{q}}_V^k = \mathbf{B}_V \hat{\mathbf{u}}_V^k. \quad (16)$$

Note that for a 1D chirp signal $\mathbf{q}[n] = \eta e^{-j n^2 \gamma d^2}$, we have $\frac{\mathbf{q}[n+1] \mathbf{q}[n-1]}{\mathbf{q}[n]^2} = e^{-j 2 \gamma d^2}$. We use this property to eliminate the effect of the scaling ambiguity. By defining

$$\rho_x^k[n_x] = \frac{\hat{\mathbf{q}}_x^k[n_x + 1] \hat{\mathbf{q}}_x^k[n_x - 1]}{\hat{\mathbf{q}}_x^k[n_x]^2} \in \mathbb{C}^{N_x - 2}, \quad (17)$$

for $n_x \in \{-\frac{N_x-1}{2} + 1, \dots, \frac{N_x-1}{2} - 1\}$ with $x \in \{H, V\}$, γ_k can be estimated by solving the following least squares (LS) problem

$$\hat{\gamma}_k = \arg \min_{\gamma} \sum_{x \in \{H, V\}} \|\rho_x^k - e^{-j 2 \gamma d_x^2} \mathbf{1}_{N_x-2}\|_2^2. \quad (18)$$

Finally, the range estimate is given by $\hat{r}_k = \frac{\pi}{\lambda \hat{\gamma}_k}$.

3) Path gain estimation: With $\{\hat{\psi}_k, \hat{\theta}_k, \hat{r}_k\}_{k=1}^K$, the RIS response is recovered as $\hat{\mathbf{b}}_k = \mathbf{b}(\hat{\psi}_k, \hat{\theta}_k, \hat{r}_k)$. Inserting $\{\hat{\mathbf{b}}_k\}_{k=1}^K$

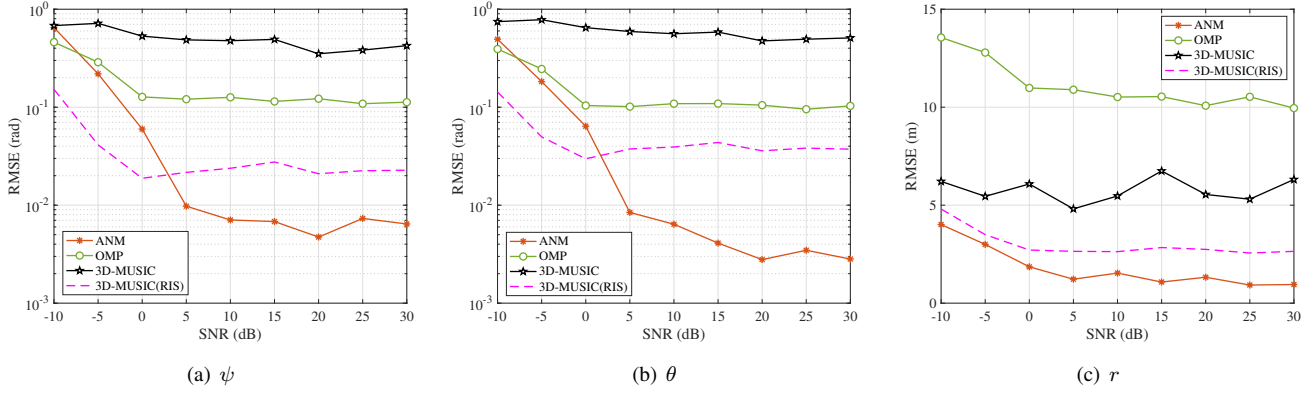


Fig. 1. RMSE performance of the azimuth-elevation-range estimation with respect to different SNR levels.

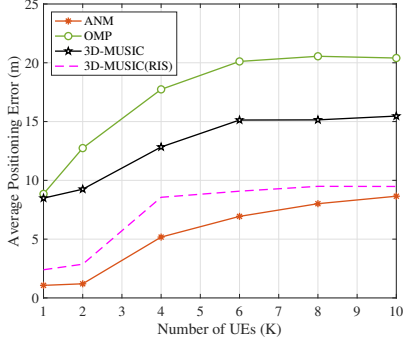


Fig. 2. Average positioning error versus the number of UEs.

into (10), the effective gains $[\tilde{\eta}_1, \dots, \tilde{\eta}_K]^T$ are obtained from least squares

$$[\hat{\tilde{\eta}}_1, \dots, \hat{\tilde{\eta}}_K]^T = (\hat{\mathbf{S}}^H \hat{\mathbf{S}})^{-1} \hat{\mathbf{S}}^H \mathbf{y}, \quad (19)$$

where $\hat{\mathbf{S}} = \bar{\mathbf{H}}[\hat{\mathbf{b}}_1, \dots, \hat{\mathbf{b}}_K] \in \mathbb{C}^{ML \times K}$. If the power p_k is known, then the path gain is estimated by $\hat{\eta}_k = \hat{\tilde{\eta}}_k / \sqrt{p_k}$.

V. SIMULATION RESULTS

We conduct numerical simulations to evaluate the proposed super-resolution method in a RIS-assisted localization scenario with $N_H = N_V = 15$, $M = 15$, and the carrier frequency $f_c = 1$ GHz, i.e., $\lambda = 0.3$ m. The inter-element spacings are $d_H = d_V = \lambda/2$. The RIS-BS link follows a free-space path-loss model with a separation of 6 m, which is assumed to be known. Each UE transmits $L = 10$ pilot symbols. The RIS reflection phases are randomly chosen between 0 and 2π . In the proposed method, subspace matrices $\mathbf{B}_H$ and $\mathbf{B}_V$ are constructed by first discretizing the quadratic phase and then applying the principal component analysis (PCA) [16], [17] to obtain a low-dimensional subspace. We choose $J_H = J_V = 3$, which is empirically found to provide an accurate approximation of the quadratic-phase chirp components. We compare the proposed algorithm against three methods: (i) the OMP algorithm with a polar-domain dictionary [10], (ii) the 3D-MUSIC algorithm [9], (iii) an idealized 3D-MUSIC that assumes direct access to RIS-side data (“3D-MUSIC(RIS)”), which serves as a lower bound for the subspace methods but

is not physically realizable due to the RIS’s reflection-only operation. All results are averaged over 50 Monte Carlo trials.

Fig. 1 shows the root mean square errors (RMSEs) of the azimuth-elevation-range estimates with respect to varying SNRs. In this experiment, $K = 2$ UEs are randomly distributed with $\psi \in [-\pi/3, \pi/3]$, $\theta \in [-\pi/6, \pi/6]$, and $r \in [3\text{m}, 15\text{m}]$, ensuring operation in the radiating near field. The results show that, as the SNR increases, the RMSEs of the proposed method decrease significantly, whereas those of the other three methods remain nearly flat for SNRs exceeding 0 dB. This saturation is primarily due to model mismatch in OMP and finite-snapshot effects in 3D-MUSIC, which limit further accuracy improvements at high SNR. Specifically, for azimuth and elevation estimation, the proposed method outperforms OMP and 3D-MUSIC when the SNR is greater than -5 dB and surpasses 3D-MUSIC(RIS) when the SNR exceeds 0 dB. For range estimation, our method consistently attains the lowest RMSE across all SNRs.

Fig. 2 shows the average positioning error $\sqrt{1/K \sum_{k=1}^K \|\mathbf{p}_k - \hat{\mathbf{p}}_k\|_2^2}$ versus the number of UEs at a fixed SNR of 15 dB, where $\hat{\mathbf{p}}_k = \hat{r}_k(\cos(\hat{\psi}_k) \cos(\hat{\theta}_k), \sin(\hat{\psi}_k) \cos(\hat{\theta}_k), \sin(\hat{\theta}_k))$ is the estimated Cartesian coordinate of the k th UE. The proposed method consistently outperforms the other methods as K increases, indicating robustness in multi-user scenarios with densely located UEs.

VI. CONCLUSIONS

This work presents a high-resolution parameter estimation method for RIS-assisted localization in the radiating NF region. By exploiting the Fresnel approximation, the RIS response admits a Kronecker-structured representation as the element-wise product of a 2D far-field steering vector and a range-dependent quadratic-phase chirp, which enables a gridless 2D super-resolution formulation for jointly estimating azimuth, elevation, and range. Simulations show that the proposed method achieves superior parameter estimation performance over representative subspace and CS methods. Future work includes developing faster solvers tailored to our problem and optimizing RIS phase patterns to further improve accuracy and robustness.

REFERENCES

- [1] M. Di Renzo, A. Zappone, M. Debbah, M.-S. Alouini, C. Yuen, J. de Rosny, and S. Tretjakov, "Smart radio environments empowered by reconfigurable intelligent surfaces: How it works, state of research, and the road ahead," *IEEE Journal on Selected Areas in Communications*, vol. 38, no. 11, pp. 2450–2525, 2020.
- [2] E. Björnson, H. Wymeersch, B. Matthiesen, P. Popovski, L. Sanguinetti, and E. de Carvalho, "Reconfigurable intelligent surfaces: A signal processing perspective with wireless applications," *IEEE Signal Processing Magazine*, vol. 39, no. 2, pp. 135–158, 2022.
- [3] H. Wymeersch, J. He, B. Denis, A. Clemente, and M. Juntti, "Radio localization and mapping with reconfigurable intelligent surfaces: Challenges, opportunities, and research directions," *IEEE Vehicular Technology Magazine*, vol. 15, no. 4, pp. 52–61, 2020.
- [4] Y. Pan, C. Pan, S. Jin, and J. Wang, "RIS-aided near-field localization and channel estimation for the terahertz system," *IEEE Journal of Selected Topics in Signal Processing*, vol. 17, no. 4, pp. 878–892, 2023.
- [5] H. Yan, H. Chen, W. Liu, S. Yang, G. Wang, and C. Yuen, "RIS-enabled joint near-field 3D localization and synchronization in isio multipath environments," *IEEE Transactions on Green Communications and Networking*, vol. 9, no. 1, pp. 367–379, 2025.
- [6] F. Liu, Y. Cui, C. Masouros, J. Xu, T. X. Han, Y. C. Eldar, and S. Buzzi, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE Journal on Selected Areas in Communications*, vol. 40, no. 6, pp. 1728–1767, 2022.
- [7] P. Stoica and R. Moses, *Spectral Analysis of Signals*. Pearson Prentice Hall, 2005.
- [8] D. Malioutov, M. Cetin, and A. Willsky, "A sparse signal reconstruction perspective for source localization with sensor arrays," *IEEE Transactions on Signal Processing*, vol. 53, no. 8, pp. 3010–3022, 2005.
- [9] P. Ramezani, A. Kosasih, and E. Björnson, "An efficient modified MUSIC algorithm for RIS-assisted near-field localization," in *GLOBECOM 2024 - 2024 IEEE Global Communications Conference*, 2024, pp. 4430–4435.
- [10] M. Cui and L. Dai, "Channel estimation for extremely large-scale MIMO: Far-field or near-field?" *IEEE Transactions on Communications*, vol. 70, no. 4, pp. 2663–2677, 2022.
- [11] X. Wei, L. Dai, Y. Zhao, G. Yu, and X. Duan, "Codebook design and beam training for extremely large-scale RIS: Far-field or near-field?" *China Communications*, vol. 19, no. 6, pp. 193–204, 2022.
- [12] S. Liu, X. Yu, Z. Gao, and D. W. K. Ng, "DPSS-based codebook design for near-field XL-MIMO channel estimation," in *ICC 2024 - IEEE International Conference on Communications*, 2024, pp. 3864–3870.
- [13] D. Yang, G. Tang, and M. B. Wakin, "Super-resolution of complex exponentials from modulations with unknown waveforms," *IEEE Transactions on Information Theory*, vol. 62, no. 10, pp. 5809–5830, 2016.
- [14] Y. Chi and Y. Chen, "Compressive two-dimensional harmonic retrieval via atomic norm minimization," *IEEE Transactions on Signal Processing*, vol. 63, no. 4, pp. 1030–1042, 2015.
- [15] M. A. Suliman and W. Dai, "Blind two-dimensional super-resolution and its performance guarantee," *IEEE Transactions on Signal Processing*, vol. 70, pp. 2844–2858, 2022.
- [16] J. Li, Q. Chen, and F. Xi, "A blind super-resolution method for near-field channel estimation with angle-range recovery," in *ICASSP 2025 - 2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2025, pp. 1–5.
- [17] F. Xi and D. Yang, "Near-field channel estimation and joint angle-range recovery in XL-MIMO systems: A gridless super-resolution approach," 2025. [Online]. Available: <https://arxiv.org/abs/2511.23187>
- [18] D. Slepian, "Prolate spheroidal wave functions, Fourier analysis, and uncertainty-v: the discrete case," *The Bell System Technical Journal*, vol. 57, no. 5, pp. 1371–1430, 1978.
- [19] Z. Zhu and M. B. Wakin, "Approximating sampled sinusoids and multiband signals using multiband modulated DPSS dictionaries," *Journal of Fourier Analysis and Applications*, vol. 23, p. 1263–1310, Dec. 2017.
- [20] M. Grant and S. Boyd, "CVX: Matlab software for disciplined convex programming, version 2.1," <https://cvxr.com/cvx>, Mar. 2014.
- [21] L. Vandenberghe and S. Boyd, "Semidefinite programming," *SIAM Review*, vol. 38, no. 1, pp. 49–95, 1996.
- [22] Y. Ran and W. Dai, "Fast and robust ADMM for blind super-resolution," in *ICASSP 2021 - 2021 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2021, pp. 5150–5154.
- [23] R. Li and D. Cabric, "A coordinate descent approach to atomic norm denoising," *IEEE Transactions on Signal Processing*, vol. 72, pp. 5077–5090, 2024.
- [24] Z. Yang, L. Xie, and P. Stoica, "Vandermonde decomposition of multilevel Toeplitz matrices with application to multidimensional super-resolution," *IEEE Transactions on Information Theory*, vol. 62, no. 6, pp. 3685–3701, 2016.